Dynamic Pedestrian Flow Guidance through Multi-robot Interaction along Human Crowd Boundaries

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Abstract

This study proposes a boundary-centric multi-robot pedestrian flow guidance design that enables soft, human-centered flow alignment along a guidance line with only a few robots. Flow boundaries are estimated from the crowd's outer contour via a convex hull, and boundary pedestrians are classified as head, side, and corner. Guidance priority integrates boundary-type importance with the predicted time to intersect the guidance line. A reachability-aware score enables optimal robot allocation via the Hungarian algorithm. Assigned robots employ a parallel-interaction strategy with a side-offset distance perpendicular to a target's walking direction to elicit anticipatory avoidance without overtly forcing heading changes. A unidirectional guidance simulation with 15 pedestrians and three robots demonstrates feasibility, showing reachability-driven role sharing, improved alignment, and guidance without pedestrians crossing the guidance line under the tested condition. Future work will evaluate safety, trust, and acceptability in opposing and crossing flows with real robots.

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods**; • **Computer systems organization** → *Robotics*; • **Computing methodologies** → *Multi-agent systems.*

Keywords

Multi-robot systems, Pedestrian flow guidance, Crowd-robot interaction, Mobile robot

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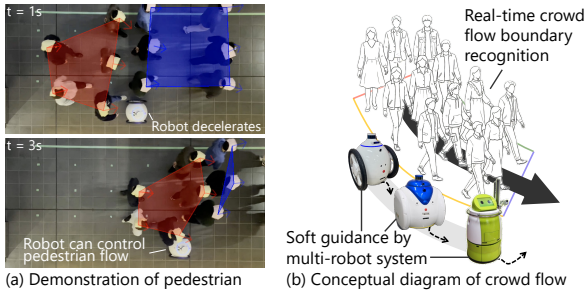
1 Introduction

Crowd management in large-scale events, transport hubs, and urban intersections is essential for safety, comfort, and mobility efficiency. When pedestrian flows are poorly regulated, participation in social and economic activities may be discouraged. Real-time, situation-adaptive interventions are therefore needed to support a safe and inclusive mobility environment.

The effectiveness of environmental design and informational interventions, such as wall layouts and signage, has been demonstrated in prior work [1, 2, 4, 11, 13, 14]. However, static infrastructure cannot readily adapt to rapid, moment-to-moment changes in pedestrian flow dynamics. Mobile robots are thus expected to complement these measures by flexibly changing their positions and roles to provide adaptive guidance.

Robot-based pedestrian guidance has often been formulated as macroscopic crowd control that models a crowd as density or velocity fields and regulates overall flow patterns [5, 8, 10, 12]. Yet macroscopic outcomes emerge from microscopic crowd–robot interaction. Boundary-focused approaches have attracted growing attention as a practical design bridge between these perspectives. Intervening with pedestrians on a dynamic pedestrian flow boundary suggests that even a small number of robots can influence not only local avoidance but also the overall shape and stability of pedestrian flow [3, 7, 9].

Our research group has shown that a single robot can influence pedestrian flow by modulating local speed and overtaking (Fig. 1(a)), but such deceleration-based intervention may be perceived as explicit or coercive. Since guidance in HRI must prioritise user experience—including safety, trust, comfort, and acceptability—we shift to soft, human-centered multi-robot guidance. Despite the promise of boundary-based control, two questions remain open in dynamic pedestrian flows: which boundary regions should be prioritised and how multiple robots should be assigned as the flow evolves. To address these questions, we propose a multi-robot pedestrian flow-boundary guidance framework that aligns pedestrian



(a) Demonstration of pedestrian flow control via a mobile robot.
(b) Conceptual diagram of crowd flow guidance via multi-robot interaction.

Figure 1: From (a) local single-robot flow control to (b) coordinated soft guidance via boundary recognition.

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flow with a predefined guidance line using a small number of robots (Fig. 1(b)). The framework combines dynamic boundary recognition with situation-adaptive role allocation, and employs a parallel-interaction strategy with a lateral side-offset to elicit anticipatory avoidance without overtly forcing heading changes.

The contributions are as follows: (1) We introduce a geometric method that estimates pedestrian flow boundaries via convex-hull approximation and identifies head/side/corner boundary pedestrians, enabling targeted intervention with a small robot team. (2) We propose a unified guidance-priority model that combines boundary-type importance with predicted time-to-intersection with a predefined guidance line, quantifying when and where guidance should be applied. (3) We develop a reachability-aware multi-robot assignment that uses the Hungarian algorithm to allocate robots to high-priority boundary targets, avoiding redundant selection and enabling situation-dependent role sharing.

2 Multi-Robot Pedestrian Flow Guidance System based on Recognized Flow Boundaries

This section details the proposed multi-robot guidance framework. First, the categorization of boundary pedestrians (head, side, corner) based on flow geometry is defined. Next, a guidance priority model that integrates boundary importance with intersection urgency is introduced. Finally, a reachability-aware assignment algorithm is described, which dynamically allocates robots to high-priority targets to achieve effective flow alignment.

2.1 System Overview

A multi-robot pedestrian flow guidance system is proposed that leverages dynamically recognized pedestrian flow boundaries (Fig. 2). A boundary pedestrian is defined as an individual on the outer contour of a pedestrian flow who is likely to influence its shape and stability. To enable effective intervention with a small number of robots, boundary pedestrians are categorised as head, side, and corner (detailed in Section 2.2). Head pedestrians are likely to shape the heading of nearby followers; side pedestrians constrain lateral spread; and corner pedestrians often act as trigger regions for local heading changes and boundary deformation.

Because the number of boundary candidates can exceed the number of robots, a guidance priority is assigned to each target by integrating boundary-type importance with the predicted time to intersect the guidance line. Multiple robots are then dynamically allocated by additionally incorporating each robot's reachability to high-priority boundary pedestrians.

Assigned robots employ a parallel-interaction strategy. Each robot moves to a position with a side-offset distance d_{side} from a target pedestrian's walking direction, on the side opposite to the desired guidance direction. This side-offset design aims to realise soft, human-centered guidance by shaping anticipatory avoidance through subtle spatial cues rather than explicit blocking or forced heading changes. This positioning is expected to elicit local avoidance and anticipation that rotate the target heading toward the guidance-line side.

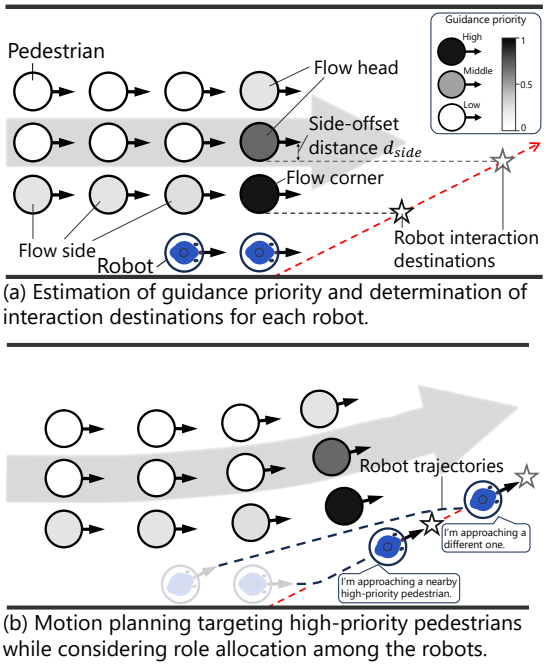


Figure 2: Schematic of the proposed multi-robot crowd guidance system.

2.2 Dynamic Pedestrian Flow Boundary Recognition

Dynamic pedestrian flow boundaries are extracted and classified using a geometric method (Fig. 3). Given a flow-specific pedestrian, its convex hull is computed to approximate the outer contour of the pedestrian flow. The flow centroid and average motion direction are estimated from this contour. For each hull edge, the boundary type is determined by evaluating the angle between its outward normal vector and the flow's average direction.

Edges with angles smaller than 45° are classified as head boundaries, those larger than 135° as tail boundaries, and those in between as side boundaries. For side boundaries, left and right are distinguished using the sign of the cross product between the average direction vector and the vector from the centroid toward each side edge: a positive sign indicates the left side and a negative sign indicates the right side.

Pedestrians within a distance d_{nb} from detected head edges are collected as $P_{\text{head}}(t)$, and those near side edges as $P_{\text{side}}(t)$. Corner pedestrians are defined as the overlap of these sets:

$$P_{\text{corner}}(t) = P_{\text{head}}(t) \cap P_{\text{side}}(t). \tag{1}$$

Head, side, and corner pedestrians form the guidance candidates because they are expected to exert strong influence on pedestrian flow shaping.

3 Dynamic Multi-Robot Assignment for Boundary Guidance

This section presents a dynamic multi-robot assignment method that allocates robots to boundary guidance targets based on the

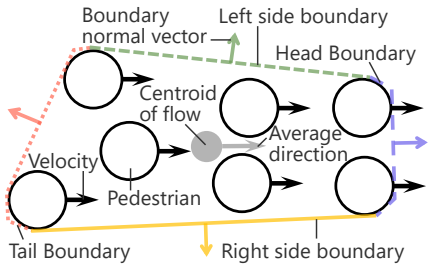


Figure 3: Schematic of pedestrian flow boundary classification. Each boundary is classified as Head ($< 45°$), Side ($45°$–$135°$), or Tail ($> 135°$) based on the angle between the edge's outward normal vector and the average flow direction.

recognized flow boundaries in Section 2.2. Flow-head, flow-side, and flow-corner pedestrians are treated as primary guidance candidates. Flow-corner pedestrians are prioritised because they are prone to trigger local heading changes and boundary deformation near the guidance line.

At each assignment cycle, the urgency of each boundary pedestrian is estimated based on whether its current motion is expected to intersect the guidance line. Let $d_j^{\perp}$ be the shortest distance from pedestrian j to the guidance line, and $v_j^{\perp}$ the signed velocity component along the unit normal pointing toward the line. The time-to-intersection is approximated as

$$t_j^{\text{line}} = \begin{cases} \dfrac{d_j^{\perp}}{v_j^{\perp}} & (v_j^{\perp} > 0), \\ +\infty & (v_j^{\perp} \le 0), \end{cases} \tag{2}$$

where $v_j^{\perp} \le 0$ indicates that pedestrian j is not moving toward the line and is therefore not expected to intersect it under the straight-motion approximation.

A base importance is assigned according to boundary type:

$$P_j^{\text{type}} = \begin{cases} 5 & j \in P_{\text{corner}}(t), \\ 2 & j \in P_{\text{head}}(t) \cup P_{\text{side}}(t). \end{cases} \tag{3}$$

These values are heuristic weights reflecting the expected influence of each boundary type; sensitivity analysis is left for future work.

A time-urgency term is defined as a monotonically decreasing function:

$$U_j(t_j^{\text{line}}) = \exp\left(-\frac{t_j^{\text{line}}}{\tau}\right), \tag{4}$$

with τ as a time-scale parameter. The unified guidance priority is then given by

$$P_j^{\text{guide}} = P_j^{\text{type}} \cdot U_j(t_j^{\text{line}}). \tag{5}$$

Given the parallel-interaction model, each target j yields a robot interaction destination $\mathbf{g}_j$. Let $\mathbf{r}_i(t)$ be the position of robot i and $v_i^{\max}$ its maximum speed. The earliest arrival time of robot i to $\mathbf{g}_j$ is estimated by

$$d_{ij} = \|\mathbf{g}_j - \mathbf{r}_i(t)\|, \qquad t_{ij}^{\text{robot}} = \frac{d_{ij}}{v_i^{\max}}. \tag{6}$$

A reachability term is defined as

$$A_{ij} = \exp\left(-\frac{t_{ij}^{\text{robot}}}{\tau_r}\right), \tag{7}$$

where τ_r is a scaling constant. The robot–target score is finally defined as

$$S_{ij} = P_j^{\text{guide}} \cdot A_{ij}. \tag{8}$$

At each cycle, robots are assigned to targets using the Hungarian algorithm to obtain the maximum-score one-to-one matching for S_{ij}, thereby avoiding duplicate target selection and ensuring an optimal division of labour.

4 Simulation-based Verification of Multi-Robot Boundary Guidance

This section presents a simulation-based proof-of-concept of the proposed flow-boundary guidance framework and the priority- and reachability-aware multi-robot assignment. A unidirectional setting is used to verify how boundary typing, priority modelling, and dynamic assignment shape pedestrian flow alignment and guidance outcomes relative to the guidance line.

Two aspects are examined: (1) whether local boundary interactions focused on high-priority targets promote alignment along the guidance line; and (2) whether these interactions enable guidance without pedestrians crossing the guidance line under the tested condition.

4.1 Unidirectional Simulation Setup

The environment contains 15 pedestrians and three mobile robots in a unidirectional flow. Pedestrian motion is generated using a standard social-force-based model [6] with a desired walking speed of 1.0 m/s. A predefined guidance line is placed as a reference for pedestrian flow alignment. The objective is to guide pedestrians to advance along the guidance-line direction and to examine whether they can be guided without crossing the guidance line under the tested condition.

4.2 Simulation Results

The results are summarised in Fig. 4. As shown in Fig. 4(a), robots preferentially approach boundary pedestrians with high guidance priority (indicated by darker shades) and determine interaction destinations based on the parallel-interaction strategy. Reachability-aware scoring induces situation-dependent role sharing, ensuring robots target reachable high-priority pedestrians.

Fig. 4(b) demonstrates the temporal reconfiguration of boundary-guidance roles. The trajectories highlight a specific behavior where a robot transitions from intervening at the flow head to supporting the flow side, adapting to the evolving crowd shape.

Fig. 4(c) shows the final pedestrian trajectories aligned with the guidance line. Under the tested unidirectional condition, pedestrians were guided without crossing the guidance line during the evaluation interval. This result indicates that priority- and reachability-aware boundary interventions can concentrate limited robots at effective boundary targets to achieve stable flow alignment.

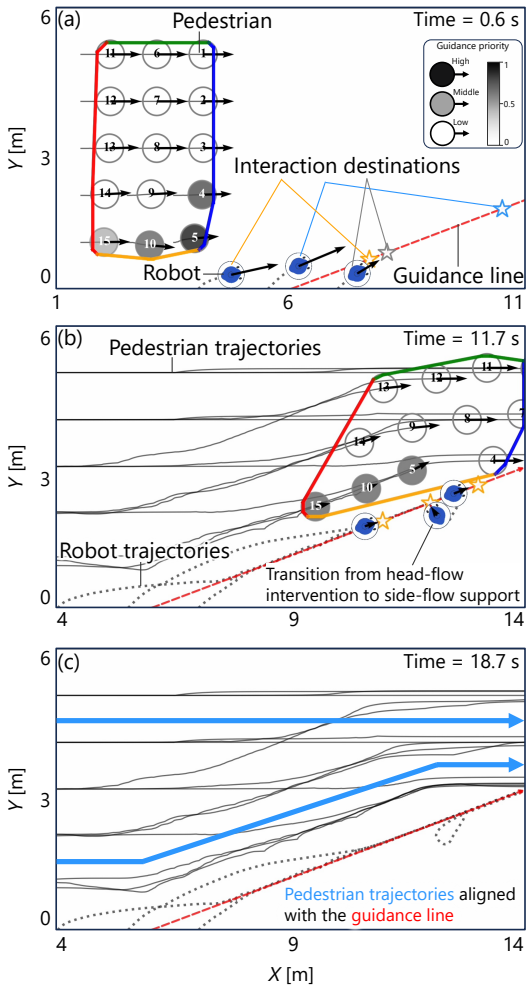


Figure 4: Results of the unidirectional pedestrian flow guidance simulation.

5 Conclusions

This paper advanced HRI-oriented pedestrian flow guidance by introducing a soft, boundary-centric multi-robot design that reduces reliance on coercive intervention. Motivated by the need for real-time, situation-adaptive support beyond static infrastructure, we proposed a multi-robot pedestrian flow-boundary guidance framework that aligns pedestrian flow with a predefined guidance line using a small number of robots. The framework detects dynamic pedestrian flow boundaries via convex-hull approximation, identifies head/side/corner boundary pedestrians relative to the mean flow direction, and computes guidance priority by integrating boundary-type importance with predicted time to intersect the guidance line. A reachability-aware score then enables optimal one-to-one robot allocation to high-priority boundary targets using the Hungarian algorithm. Assigned robots employ a parallel-interaction strategy with a lateral side-offset to elicit anticipatory avoidance and gentle heading adjustment, linking the control mechanism to human-centered guidance design.

A unidirectional guidance simulation with 15 pedestrians and three robots provided a feasibility demonstration. The results showed reachability-driven role sharing and temporal reconfiguration of boundary roles, improved alignment with the guidance line, and guidance without pedestrians crossing the guidance line under the tested condition. These findings support the usefulness of boundary typing and reachability-aware assignment as a coherent design mechanism for soft multi-robot pedestrian flow guidance.

Future work will extend evaluations to complex scenarios like opposing and crossing flows, and quantify trajectory smoothness to validate the human-centered nature of the approach compared to static obstacles. We also plan to analyze sensitivity to the side-offset distance, hypothesizing that this parameter tunes the intervention between coercive blocking and soft guidance. Finally, although the optimal robot-to-crowd ratio remains an open question, we hypothesize that appropriate HRI designs—such as audio, gaze, or light signals—can foster crowd cooperation to maximize guidance effectiveness. Future experiments will verify this hypothesis and define the necessary design requirements.

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